MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, OIL, HELICOPTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a voltage ratio type, oil pressure transmitter, designed to operate within the vibration envelope peculiar to helicopters.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-58097	Transmitters, Pressure, Aircraft, Voltage Ratio Type, General Specification for
MIL-I-58098	Indicators, Aircraft, Pressure, Voltage Ratio Type, General Specification for
MIL-I-58099	Indicator, Pressure, Type TB-1

STANDARDS

Military

MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MIL-STD-831	Test Reports, Preparation of

(Copies of documents required by suppliers may be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6620

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing. The requirements of MIL-T-58097 apply to this specification except for deletions or additions specified herein. If the two specifications should conflict this specification shall govern.

3.1.1 Preproduction sample. A preproduction sample is defined as a production line sample of the product which a manufacturer will subsequently produce in fulfillment of the contract. Approval of preproduction sample shall be obtained from the procuring activity prior to production of contract items.

3.2 Design and construction.

3.2.1 The transmitter shall be so designed that it will sense the differential of two pressures and convert the differential pressure measurement into an electrical signal for use with indicators conforming to MIL-I-58098 and MIL-I-58099.

3.2.2 Attitude. The transmitter shall be of such design that it will operate as specified herein in any position.

3.2.3 Adjustments. The transmitter shall not require periodic adjustments.

3.2.4 Lead length. The transmitter shall be so designed that it will operate as specified herein when the lead length (No. 20 AWG stranded copper wire) between the transmitter and indicator is varied from 5 to 300 feet.

3.3 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.3.1 Reliability. The transmitter shall have a minimum acceptable mean-time-between-failures (MTBF) of 2,000 hours based on a discrimination ratio of 0.67 and a confidence level of 90 percent.

3.3.2 Longevity. The transmitter shall have an operating life span (equipment longevity as defined in MIL-STD-781) of 3,000 hours.

3.4 Performance.

3.4.1 Tolerances. The transmitter shall operate within the tolerances specified in Tables I and II.

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3.5 Weight. The weight of the transmitter shall not exceed 1.0 pound.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of transmitters shall be as specified herein and in MIL-T-58097.

4.3 Preproduction testing.

4.3.1 Preproduction tests. The preproduction tests shall consist of all the tests specified herein and in MIL-T-58097.

4.3.2 Test samples. The test samples shall consist of ten transmitters representative of the production equipment. Three of the samples shall be subjected to all the tests except reliability and the other seven samples shall be subjected to the reliability test only. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.3.3 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

4.4 Test methods.

4.4.1 Reliability.

4.4.1.1 Demonstration. Unless otherwise specified, seven transmitters shall be subjected to the reliability test specified in 4.4.1.2 which shall demonstrate an MTBF of 2,000 hours at a confidence level of 90 percent. The total test time, number of failures permitted, and accept, continue-test, reject criteria shall be in accordance with Figure 1.

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TABLE I. Scale Error

Test Pressure PSI	Tolerance ± PSI			
	Room Temp	Seasoning Overpressure Power Supply Humidity	Low and High Temp Low and High Temp Exposure Vibration, Endurance Altitude, Room Temp after 232°C Exposure	Extreme High Temp
0	1.0	1.75	2.0	2.5
10	1.0	1.75	2.0	2.5
20	1.0	1.75	2.75	3.25
30	1.75	2.25	2.75	3.25
40	1.75	2.25	2.75	3.25
50	1.75	2.25	2.75	3.25
60	1.75	2.25	2.75	3.25
70	1.75	2.25	2.75	3.25
80	1.75	2.25	2.75	3.25
90	2.0	2.50	3.0	3.5
100	2.0	2.50	3.0	3.5

TABLE II Friction, Position, and Vibration Error

Test	Tolerance ± PSI
Friction	
Room Temperature	0.5
High	0.5
Low	1.5
Endurance	0.5
Humidity	0.5
Position Error at 50 psi	0.5
Vibration	
Pointer Oscillation	0.5
Pointer Variation	0.5

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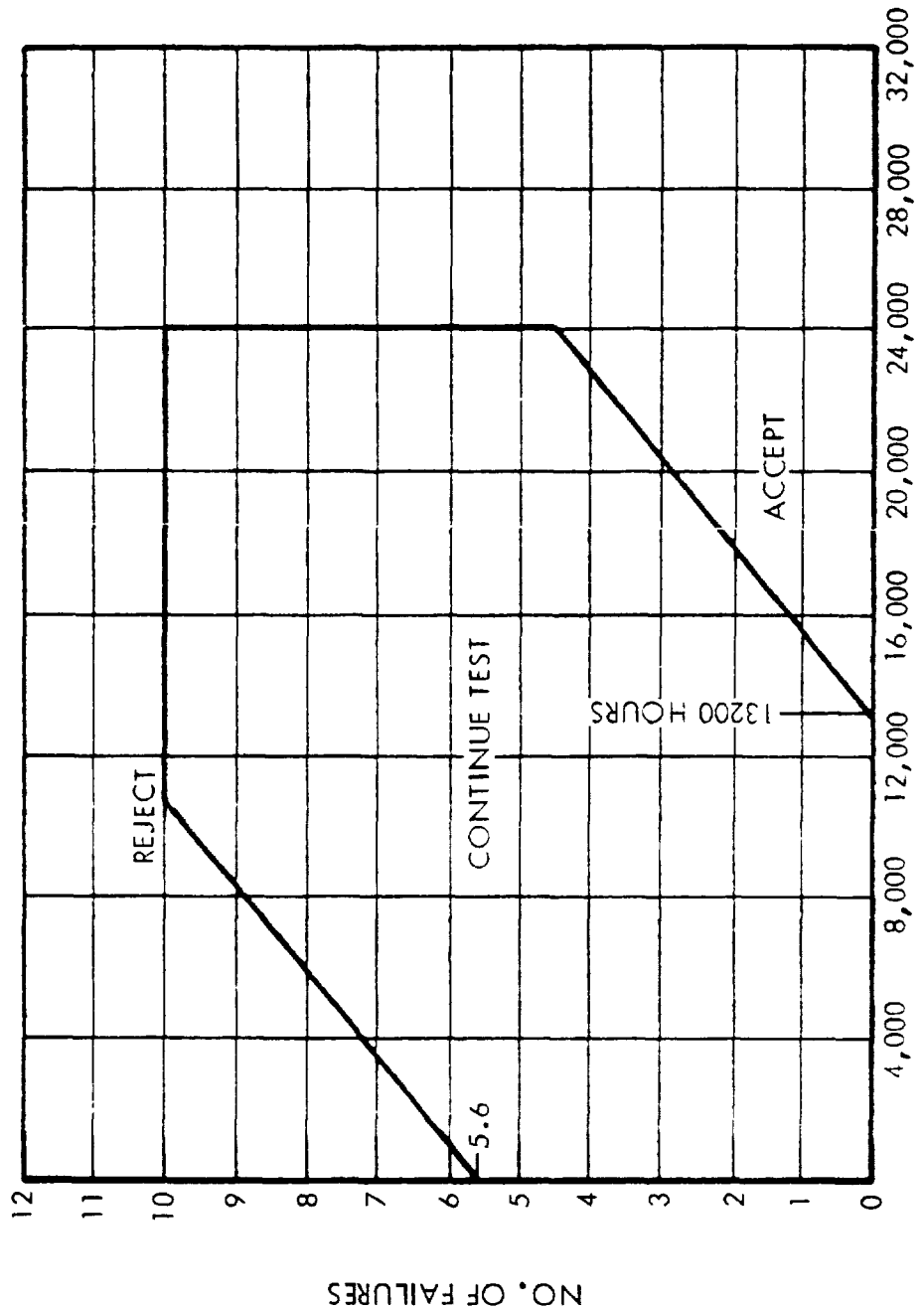


FIGURE 1. ACCEPT, CONTINUE, REJECT - TEST CRITERIA

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4.4.1.2 Test. The test equipment shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, input signal shall be applied to the transmitter at a rate of 2 to 3 cycles per minute from 0 to 85 percent of full scale and return. The test shall begin when rated power is applied. After completion of 4 hours of operation with rated power on, which shall include at least 480 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the transmitter shall be vibrated in accordance with MIL-STD-781, test level J. The temperature cycles shall be conducted in the following sequence: 4 hours at -54°C, temperature change to +165°C at a minimum rate of 5°C per minute, 4 hours at +165°C, temperature change to -54°C, and 4 hours at -54°C. In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. Upon completion of each 250 hours of cycling, the transmitter shall be checked for scale error. The scale error reading shall not exceed 150 percent of the room temperature tolerance in Table I. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report in accordance with MIL-STD-785 which shall contain test procedures, details of any malfunctions, and corrective actions taken.

4.4.2 Longevity. At the completion of the reliability test, three transmitters shall continue to be tested for a total of 3,000 hours each. The test time accumulated during the reliability test shall be applied to this test. The accept-reject criteria is not applicable.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-T-58097. All openings shall be plugged or sealed to prevent the entrance of dirt and moisture.

6 NOTES

6.1 Intended use. The transmitter covered by this specification is intended for use in helicopter oil pressure indicating systems with indicators conforming to MIL-I-58099(AV).

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6.2 Ordering data. Procurement document should specify the following:

a. Title, number, and date of this specification

b. Information, if any, to be submitted with the preproduction samples

c. When sampling plan B tests will not be conducted

d. Levels of packaging and packing required.

Custodian.

ARMY-AV

Preparing Activity

ARMY-AV
PP. NO. 6620-A006

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St Louis, MO 63166

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCO

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

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